

SECONDARY EDUCATION EXAMINATION (SEE)

Model Question 2080 — SET B (Set 3) Solutions

GROUP 'A' (Multiple Choice Questions — 10 x 1 = 10 Marks)

Question	Correct Option	Answer Details
(a) Which of the following is the function of receiver?	(i)	To convert received signal into desired form.
(b) Why are mosses and ferns both found in moist and shady places?	(iv)	Requires water for fertilization.
(c) Which statement is true for organisms A (segmented) and B (unsegmented)?	(iii)	Organism A has well developed circulatory system whereas organism B has no circulatory system.
(d) Which of the following groups is a group of endangered animals?	(i)	Black buck, Wild yak, One horned Rhinoceros
(e) If mass and radius both are made halved, what is the new value of g on the earth surface?	(ii)	2 g
(f) On which principle is the given hydraulic lift instrument based on?	(i)	Pascal's law
(g) Which is the correct order of speed of violet, blue, yellow, and red in prism?	(ii)	$v_v > v_b > v_y > v_r$
(h) Under what condition can the flat universe be hypothesized?	(i)	When average density is equal with the critical density.
(i) What is X in the given chemical equation: $\text{NaOH} + \text{X} \rightarrow \text{NaCl} + \text{H}_2\text{O}$?	(iii)	HCl
(j) Which of the following is a threat to aquatic life due to excess organic matter?	(ii)	Microorganisms consume dissolved oxygen to decompose organic matter.

GROUP 'B' (Very Short Answer Questions — 9 x 1 = 9 Marks)

(a) What is dependent variables? Write with an example.

Answer: A dependent variable is the variable being tested and measured in a scientific experiment, which changes in response to the independent variable. *Example:* The height of a plant measured over time.

(b) Write the relation between upthrust and weight of displaced liquid.

Answer: Upthrust is directly equal to the weight of the displaced liquid (Upthrust = Weight of displaced liquid).

(c) The apparent depth of water in a pond appears less than the real depth. Why?

Answer: This occurs due to the refraction of light as rays travel from the denser medium (water) to the rarer medium (air), bending away from the normal line.

(d) What is the focal length of lens having power 1 dioptre?

Answer: Since Focal Length (f) = $1 / \text{Power (P)}$, $f = 1 / 1 = 1$ meter (or 100 cm).

(e) A fan in a room rotates when its switch is made on. Which principle is used here?

Answer: It works on the magnetic effect of electric current, where electrical energy is transformed directly into mechanical energy.

(f) Which force is responsible for the existence of heavenly bodies in the universe?

Answer: Gravitational force is responsible for holding galaxies, stars, planets, and celestial structures together.

(g) Write the name of main ore of iron.

Answer: Hematite (Fe_2O_3) is the primary mineral ore of iron extraction.

(h) Why reduction of active metal is possible only by the electro refining method?

Answer: Highly active metals possess high affinity for oxygen and chemical bonds, making standard reducing agents ineffective; electricity easily isolates them via electrolysis.

(i) Write molecular formula of given compound (Propane).

Answer: The structure displays 3 Carbon atoms bound to 8 Hydrogen atoms, giving the precise formula C_3H_8 .

GROUP 'C' (Short Answer Questions — 14 x 2 = 28 Marks)

3. Prove that the unit of potential difference (V) = $\text{kg m}^2 \text{s}^{-3} \text{A}^{-1}$.**Proof:**

Potential Difference (V) = Work Done (W) / Charge (Q)

• Unit of Work Done (W) = Joules = Newtons $\times$ meters = $(\text{kg m s}^{-2}) \times \text{m} = \text{kg m}^2 \text{s}^{-2}$

• Unit of Charge (Q) = Current (I) $\times$ time (t) = Ampere $\times$ second = A s

Substituting units: $V = (\text{kg m}^2 \text{s}^{-2}) / (\text{A s}) = \text{kg m}^2 \text{s}^{-3} \text{A}^{-1}$.

Hence, the expression is derived and verified precisely.

4. Write any two characteristics of gymnosperm.**Answer:**

1. Seeds are completely naked and not enclosed inside fruits.
2. Reproductive organs are developed into reproductive cones instead of conventional true flowers.

5. Which process is represented by A and B in the honey bee concept map?**Answer:**

- **Process A:** Parthenogenesis (Development of an unfertilized haploid egg into a male Drone bee).
- **Process B:** Fertilization (Fusion of haploid sperm and egg to form a diploid female Queen/Worker bee).

6. Write any two differences between diploid cell and haploid cell.**Answer:**

1. Diploid cells contain two complete paired sets of chromosomes (2n), while haploid cells hold only a single set (n).
2. Diploid cells are produced via mitotic replication (somatic tissue), whereas haploid cells are formed by meiotic division (gamete tissue).

7. In which two conditions In-vitro fertilization (IVF) helps for reproduction in women?**Answer:**

1. When a woman has blocked, heavily damaged, or completely removed Fallopian tubes.
2. In cases of low sperm count or poor sperm motility in the male partner, preventing natural fertilization.

8. Valves are present in veins but not in arteries, why?

Answer: Veins transport blood under low structural pressure back to the heart against gravity. Inner semi-lunar valves prevent blood backflow. Arteries receive high-pressure blood pumped straight out from ventricles, making valves redundant.

9. Identify the medicinal plant shown in figure (a) [Cinchona / Neem / Aloe-vera] and write its two uses.

Answer: Based on high secondary curriculum reference sets:

1. It serves as a potent natural remedy to treat malaria and chronic fever.
2. Used in manufacturing natural organic tonics and skin antiseptic treatments.

10. Give any two examples of free fall.**Answer:**

1. A spacecraft or artificial satellite revolving around Earth in a continuous orbital path.
2. An object dropping in an artificially created complete vacuum space where no air friction exists.

11. A body is dropped from the top of the tower of height 'h' and at the same time next body is thrown up from the bottom of tower with velocity 20 m/s. If these bodies meet after 2 sec. at a point of the tower, calculate the height of the tower. ($g = 10 \text{ m/s}^2$)**Solution:**

Let the height of the meeting point from the ground be y . Then the distance traveled by the falling body from the top is $(h - y)$.

- For the dropped object ($u = 0$):

$$\text{Distance } (h - y) = u \times t + 0.5 \times g \times t^2 = 0 + 0.5 \times 10 \times (2)^2 = 20 \text{ m} \text{ — (Equation 1)}$$

- For the vertically projected object ($u = 20 \text{ m/s}$):

$$\text{Distance } y = u \times t - 0.5 \times g \times t^2 = 20 \times 2 - 0.5 \times 10 \times (2)^2 = 40 - 20 = 20 \text{ m} \text{ — (Equation 2)}$$

- Adding Equation 1 and Equation 2:

$$(h - y) + y = 20 + 20 \Rightarrow \mathbf{h = 40 \text{ meters.}}$$

12. Write an application of step-up transformer and step-down transformer each.**Answer:**

- **Step-up Transformer:** Used at power generation stations to raise electric voltage for efficient long-distance transmission.
- **Step-down Transformer:** Used near substations and inside household chargers to reduce voltage to safe levels (e.g., 220V down to 5V).

13. Write any two ways of increasing the amount of induced electromotive force (e.m.f.) in an AC generator.**Answer:**

1. Increasing the total number of turns of insulated wire inside the rotating armature coil.
2. Increasing the relative speed of rotation of the armature coil within the magnetic field.

14. Equal volume of H_2O_2 is kept in conical flask P and Q. If some MnO_2 and H_3PO_4 are added to P and Q respectively, in which flask is the reaction faster? Explain.

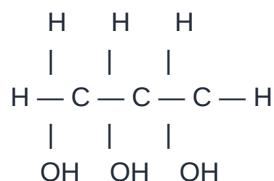
Answer: The rate of chemical reaction is significantly faster in **Conical Flask P**. Manganese dioxide (MnO_2) acts as a positive catalyst, accelerating the decomposition rate of hydrogen peroxide. In contrast, phosphoric acid (H_3PO_4) acts as a negative catalyst, slowing down the decomposition rate inside flask Q.

15. Hydrocarbon 'A' preserves biological specimens and 'B' serves as a sweetening agent. Write IUPAC name of 'A' and structural formula of 'B'.

Answer:

- **Compound A** is Formaldehyde solution. Its official IUPAC name is **Methanal**.
- **Compound B** is Glycerol. Its IUPAC name is Propane-1,2,3-triol.

Structural Formula of B (Glycerol):



16. Write any two applications of Calamus (Bojo) in daily life.

Answer:

1. It is chewed raw or consumed in extracts to cure sore throats, coughs, and throat irritations.
2. Used traditionally to relieve digestive issues, stomach cramps, and loss of appetite.

GROUP 'D' (Long Answer Questions — 7 x 4 = 28 Marks)

17. Study the communication diagram and answer: (a) Write two examples of this technology. (b) Write the functions of transmitter and receiver.

Answer:

(a) Digital Communication Technology Examples: 1. Mobile phone telecommunication networks. 2. Fiber-optic high-speed internet data routers.

(b) Functional Operations:

- **Transmitter:** Modulates, encodes, and amplifies structural source data signals, launching them cleanly into the transit medium channel.
- **Receiver:** Captures incoming weak channel signals, filtering noise and decoding them accurately back into original formats for end-users.

18. A family has only two daughters. Does it mean the husband lacks 'Y' chromosome sperms? Explain with a sex determination chart.

Answer: No, it does not mean the husband lacks 'Y' sperms. Human males produce X-bearing and Y-bearing sperms in an exact 50:50 ratio. Fertilization is a purely random statistical event.

Sex Determination Chart:

- Parents: Father (XY) × Mother (XX)
- Gametes from Father: X and Y | Gametes from Mother: X and X
- Random Cross Combinations:
 1. Sperm X + Egg X = XX (Daughter, 50% probability)
 2. Sperm Y + Egg X = XY (Son, 50% probability)

Therefore, having two daughters is a matter of chance and does not imply a biological deficiency in Y-sperm production.

19. Based on the blood composition diagram: (a) Name B. (b) Identify disease if A is low. (c) Write main function of C. (d) Write life span of B.

Answer:

- **(a) Name of B:** White Blood Cells (WBC) / Leucocytes.
- **(b) Component A Deficiency Disease:** Component A represents Red Blood Cells (RBC). A deficiency leads to **Anaemia**.
- **(c) Main Function of C:** Component C represents Blood Platelets (Thrombocytes). Their primary role is blood clotting at injury sites.
- **(d) Life Span of B:** White blood cells generally have a life span ranging from a few days up to about 2 weeks (approx. 12-20 days) depending on type.

20. Study the specific heat table: (a) Which cools fastest? (b) Which requires more heat energy to raise temperature? Explain both.

Data: A = 234 J/kg°C, B = 400 J/kg°C, C = 900 J/kg°C.

Answer:

- **(a) Fastest Cooling Component: Metal A** cools down the fastest because it has the lowest specific heat capacity. It stores less thermal energy per unit mass and releases its heat quickly.
- **(b) Highest Thermal Capacity Demand: Metal C** requires the most heat energy. Having the highest specific heat capacity means it requires more energy to disrupt internal molecular lattices and raise its temperature by 1°C.

21. A man can read books only by keeping them away from his eyes. (a) Identify the defect. (b) Write one cause. (c) Suggest corrective measures with its functional principle.

Answer:

- **(a) Identification of Visual Defect:** Long-sightedness or **Hypermetropia**.
- **(b) Primary Biological Cause:** The eyeball becomes too short along its axial path, or the crystalline lens becomes too flat, reducing its refractive power.
- **(c) Recommended Corrective Measures:** The individual must use spectacles equipped with a tailored **Convex Lens**. A convex lens converges incoming rays beforehand so that the eye's internal lens focuses them directly onto the retina instead of behind it.

22. Based on periodic table trends: (a) Identify the period. (b) Write symbols for highly reactive metal/non-metal. (c) Write inert element symbol. (d) Differentiate B and F.

Context Analysis: Elements form NaCl (vegetable salt), identifying A as Sodium (Na) and G as Chlorine (Cl). Hence, these belong to Period 3.

Answer:

- **(a) Period Category:** **Period 3** of the Modern Periodic Table.
- **(b) Reactive Symbols:** Highly reactive metal is **A (Na)**; highly reactive non-metal is **G (Cl)**.
- **(c) Inert Element Symbol:** Element **H (Argon - Ar)**, which has a complete outer valency octet.
- **(d) Difference between B (Mg) and F (S):** Element B is a highly electropositive metallic conductor that forms basic oxides, while F is an electronegative non-metal that forms acidic oxides.

23. Describe laboratory preparation of Carbon Dioxide gas, its litmus paper testing mechanism, and a precaution.

Answer:

- **Chemical Equation:** $\text{CaCO}_3 + 2\text{HCl} \rightarrow \text{CaCl}_2 + \text{H}_2\text{O} + \text{CO}_2$
- **Litmus Test Verification:** When a moist blue litmus paper is introduced into a gas jar filled with carbon dioxide, it turns red. This proves that CO_2 dissolves in water to form weak carbonic acid (H_2CO_3), demonstrating its acidic nature.
- **Precautionary Protocol:** The lower end of the thistle funnel must be fully submerged beneath the liquid acid inside the Woulf's bottle to prevent generated gas from escaping upward through the funnel.

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